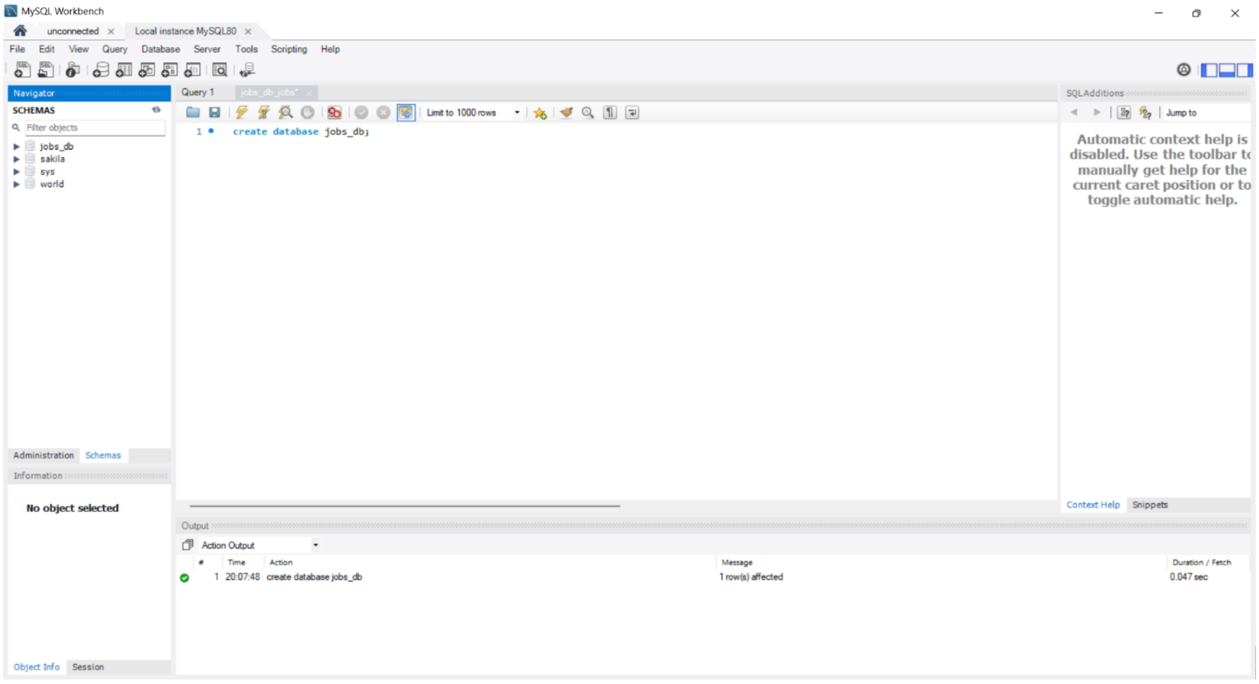
MYSQL DATA LOADING & CLEANING STEP BY STEP

1) Open Your Workbench and Create a Database with name “jobs_db” and refresh the ‘Schemas’ to see whether you database is created or not.



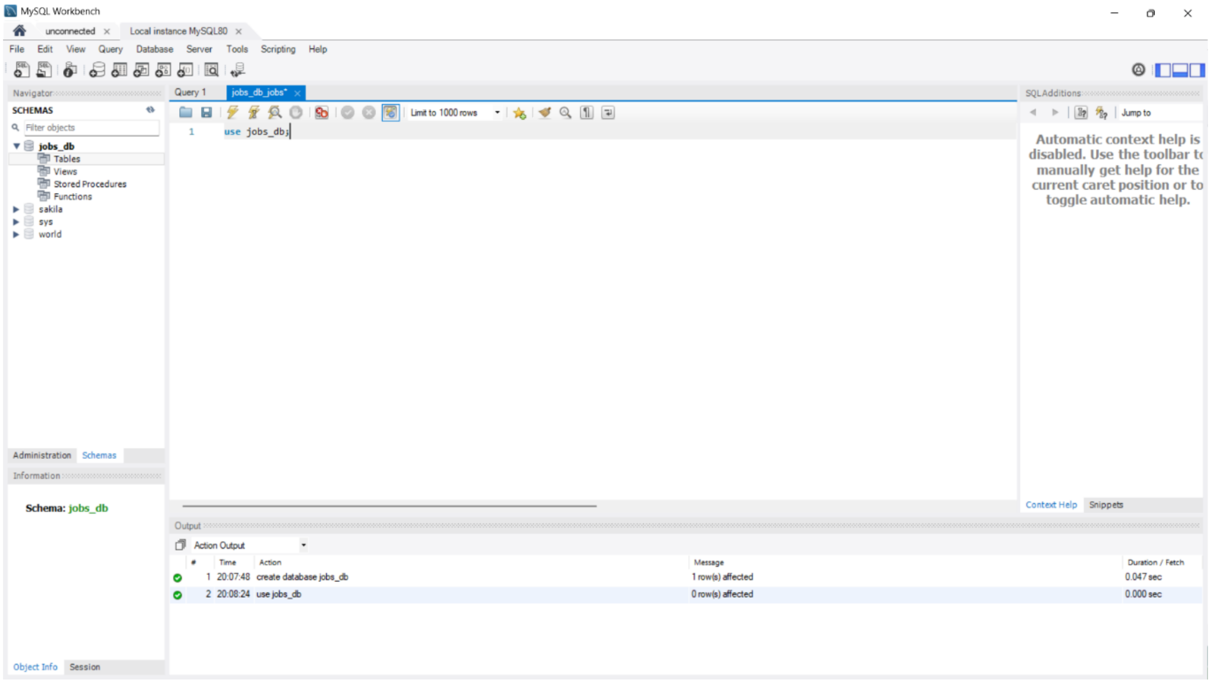
```
create database jobs_db;
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

No object selected

#	Time	Action	Message	Duration / Fetch
1	20:07:48	create database jobs_db	1 row(s) affected	0.047 sec

2) To use the database use the query “use jobs_db” this will make your database active and ready to use.



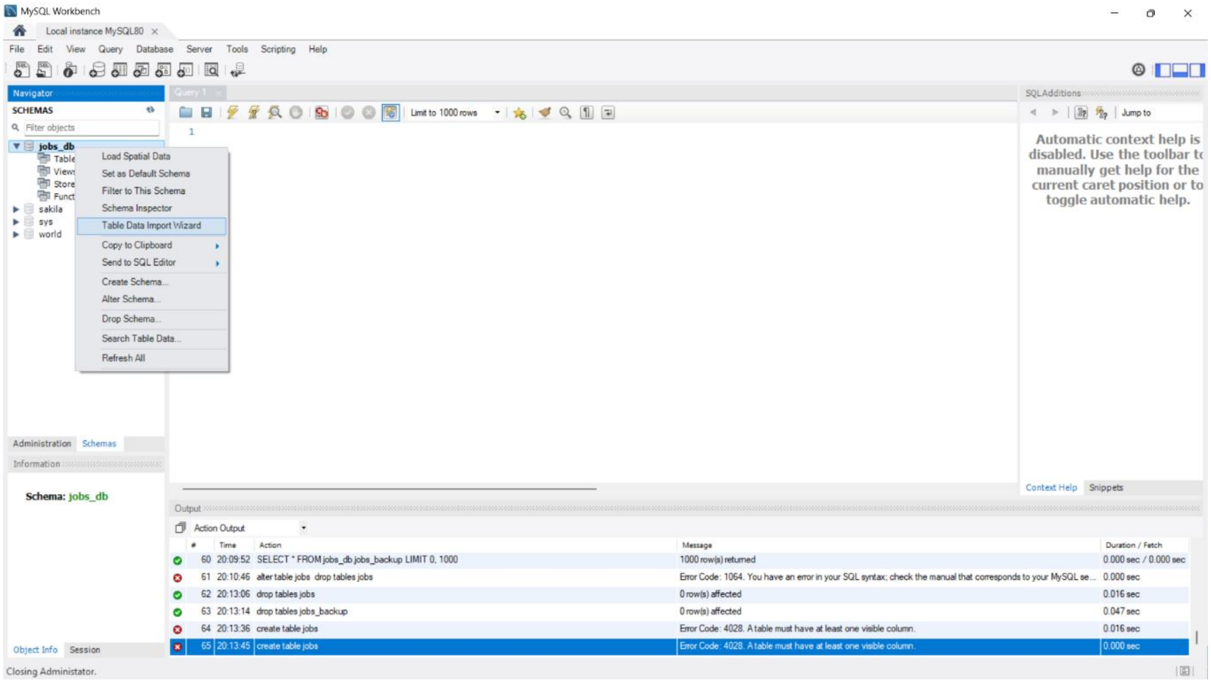
```
use jobs_db;
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Schema: jobs_db

#	Time	Action	Message	Duration / Fetch
1	20:07:48	create database jobs_db	1 row(s) affected	0.047 sec
2	20:08:24	use jobs_db	0 row(s) affected	0.000 sec

3) Right click on the database and select the option 'Table Data Import/ Wizard' to import your Raw Data.


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
jobs_db
sakila
sys
world
Load Spatial Data
Set as Default Schema
Filter to This Schema
Schema Inspector
Table Data Import Wizard
Copy to Clipboard
Send to SQL Editor
Create Schema...
Alter Schema...
Drop Schema...
Search Table Data...
Refresh All
Limit to 1000 rows
SQLAdditions
Jump to
Context Help
Snippets
Administration
Schemas
Information
Object Info
Session

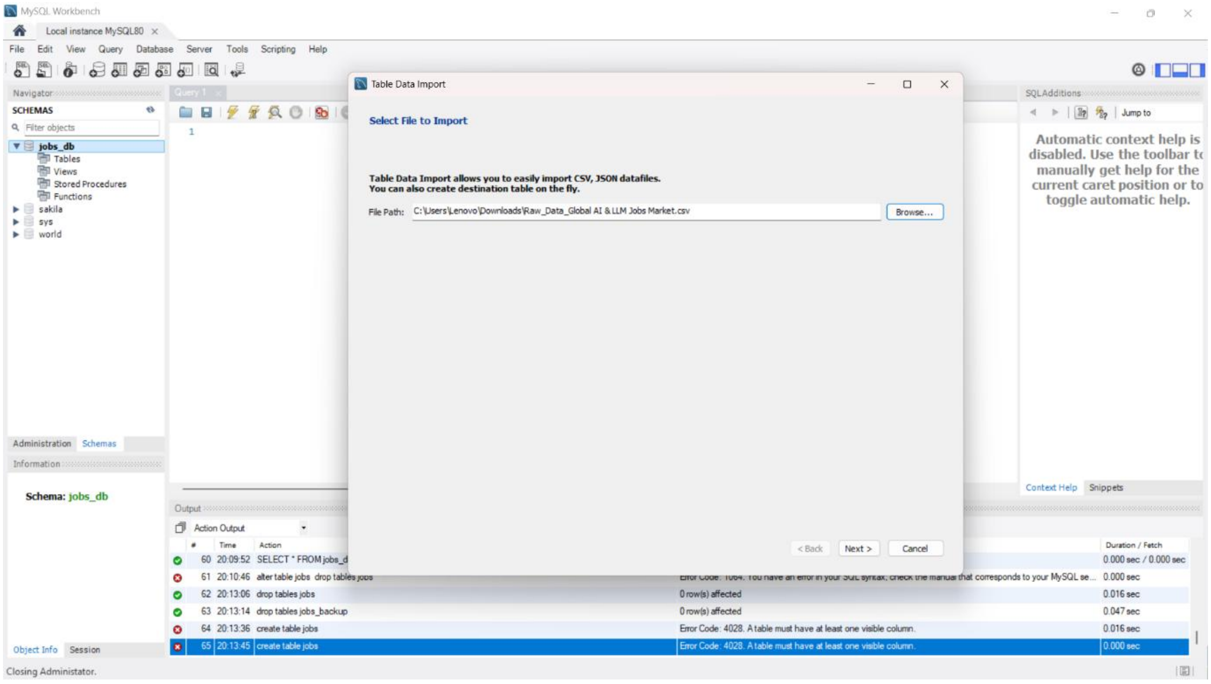
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Schema: jobs_db

```
Output
Action Output
 #   Time      Action                                        Message                                                                                 Duration / Fetch
 60  20:09:52  SELECT * FROM jobs_db.jobs_backup LIMIT 0, 1000  1000 row(s) returned                                                                  0.000 sec / 0.000 sec
 61  20:10:46  alter table jobs  drop tables jobs            Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...  0.000 sec
 62  20:13:06  drop tables jobs                              0 row(s) affected                                                                       0.016 sec
 63  20:13:14  drop tables jobs_backup                       0 row(s) affected                                                                       0.047 sec
 64  20:13:36  create table jobs                             Error Code: 4028. A table must have at least one visible column.                        0.016 sec
 65  20:13:45  create table jobs                             Error Code: 4028. A table must have at least one visible column.                        0.000 sec
```

Closing Administrator.

4) Select the file path to import the data.


MySQL Workbench
Local instance MySQL80
Table Data Import
Browse...
< Back
Next >
Cancel

Select File to Import

Table Data Import allows you to easily import CSV, JSON datafiles.
You can also create destination table on the fly.

File Path: C:\Users\Lenovo\Downloads\Raw_Data_Global AI & LLM Jobs Market.csv

5) Rename the Table like ‘jobs’ or anything as you wish.

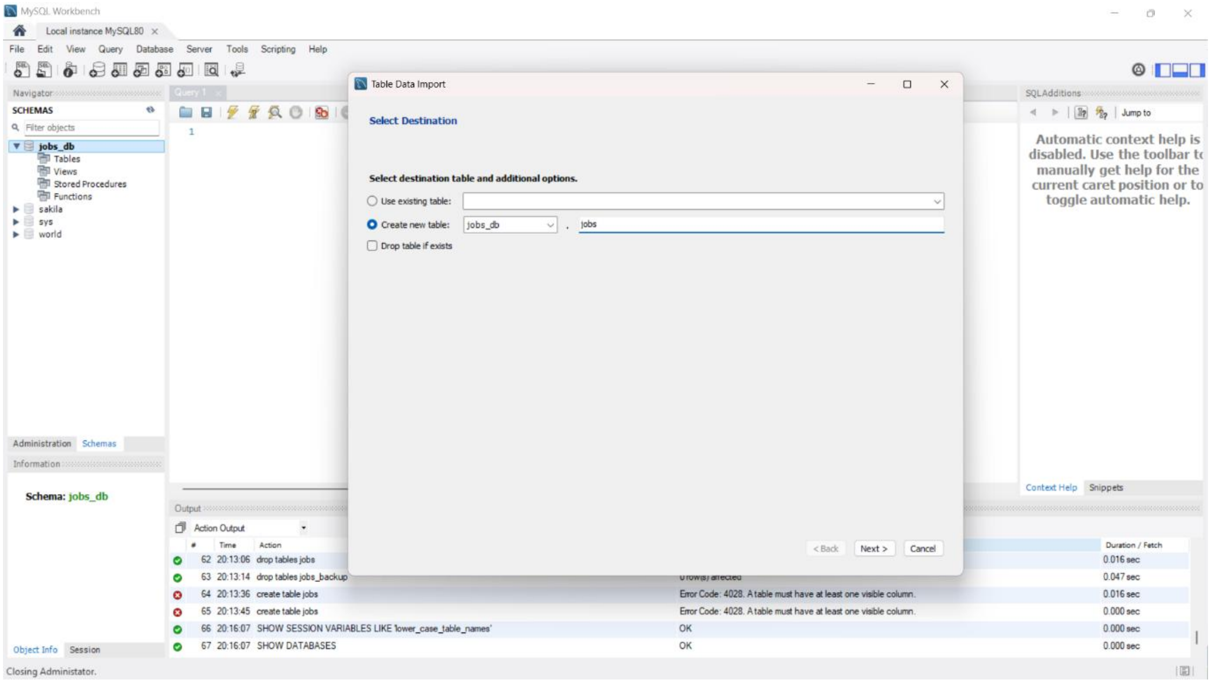


Table Data Import

Select Destination

Select destination table and additional options.

○ Use existing table:

● Create new table: jobs_db . jobs

☐ Drop table if exists

MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
jobs_db
Tables
Views
Stored Procedures
Functions
sakila
sys
world
Administration
Schemas
Information
Schema: jobs_db
Object Info
Session
Query 1
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help
Snippets
< Back
Next >
Cancel
Output
Action Output
Closing Administator.

```
#   Time      Action                                                                                                         Duration / Fetch
62  20:13:06  drop tables jobs                                                                                               0.016 sec
63  20:13:14  drop tables jobs_backup                                0 row(s) affected                                       0.047 sec
64  20:13:36  create table jobs                                      Error Code: 4028. A table must have at least one visible column.  0.016 sec
65  20:13:45  create table jobs                                      Error Code: 4028. A table must have at least one visible column.  0.000 sec
66  20:16:07  SHOW SESSION VARIABLES LIKE 'lower_case_table_names'   OK                                                      0.000 sec
67  20:16:07  SHOW DATABASES                                         OK                                                      0.000 sec
```

6) Click next if you want to change the datatype you can do but we will change the datatype later with query.

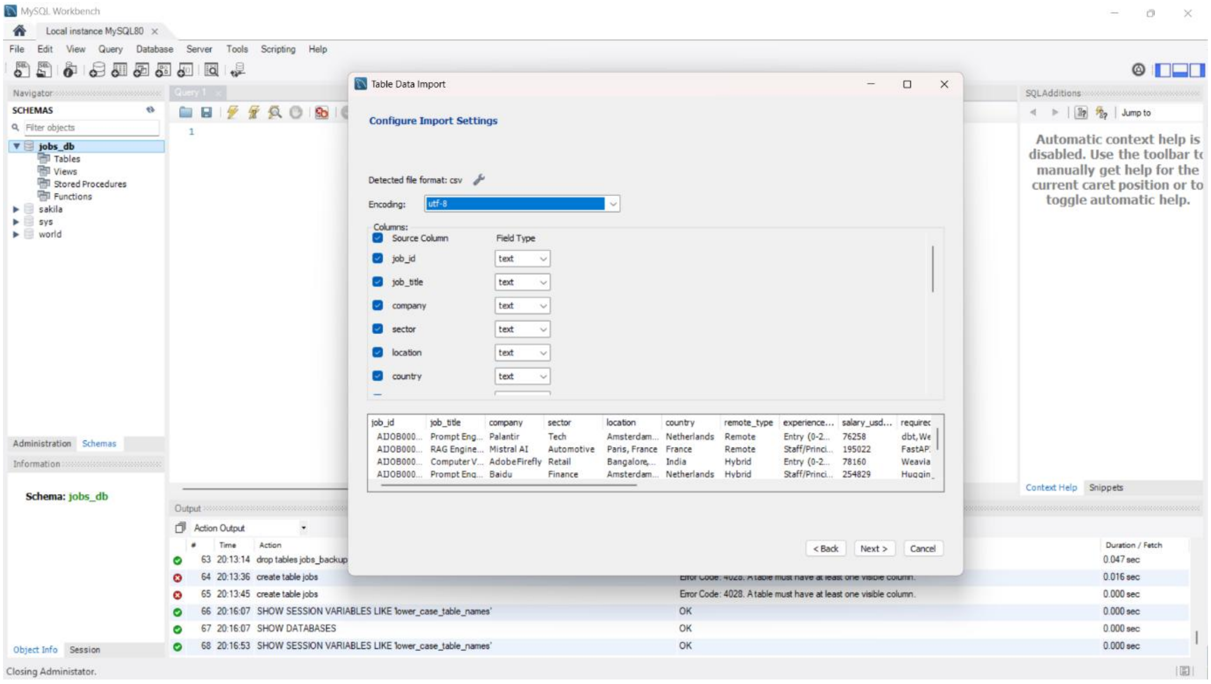


Table Data Import

Configure Import Settings

Detected file format: csv

Encoding: utf-8

Columns:

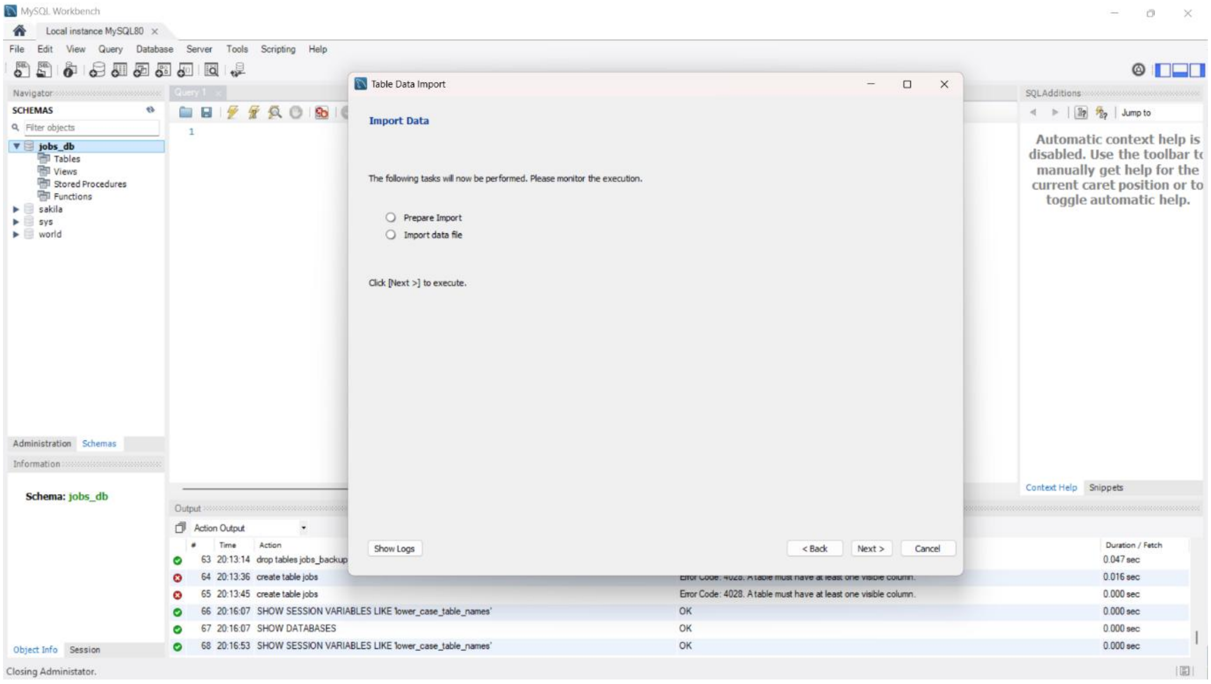
☑ Source Column	Field Type
☑ job_id	text
☑ job_title	text
☑ company	text
☑ sector	text
☑ location	text
☑ country	text

job_id	job_title	company	sector	location	country	remote_type	experience...	salary_usd...	requirec
AIJOB000...	Prompt Eng...	Palantir	Tech	Amsterdam...	Netherlands	Remote	Entry (0-2...	76258	dbt, We
AIJOB000...	RAG Engine...	Mistral AI	Automotive	Paris, France	France	Remote	Staff/Princi...	195022	FastAP:
AIJOB000...	Computer V...	AdobeFirefly	Retail	Bangalore,...	India	Hybrid	Entry (0-2...	78160	Weavia
AIJOB000...	Prompt Eng...	Baidu	Finance	Amsterdam...	Netherlands	Hybrid	Staff/Princi...	254829	Huggin_

MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
jobs_db
Tables
Views
Stored Procedures
Functions
sakila
sys
world
Administration
Schemas
Information
Schema: jobs_db
Object Info
Session
Query 1
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help
Snippets
< Back
Next >
Cancel
Output
Action Output
Closing Administator.

```
#   Time      Action                                                                                                         Duration / Fetch
63  20:13:14  drop tables jobs_backup                                                                                        0.047 sec
64  20:13:36  create table jobs                                      Error Code: 4028. A table must have at least one visible column.  0.016 sec
65  20:13:45  create table jobs                                      Error Code: 4028. A table must have at least one visible column.  0.000 sec
66  20:16:07  SHOW SESSION VARIABLES LIKE 'lower_case_table_names'   OK                                                      0.000 sec
67  20:16:07  SHOW DATABASES                                         OK                                                      0.000 sec
68  20:16:53  SHOW SESSION VARIABLES LIKE 'lower_case_table_names'   OK                                                      0.000 sec
```

7) Click next to start importing the files in the workbench.



Import Data

The following tasks will now be performed. Please monitor the execution.

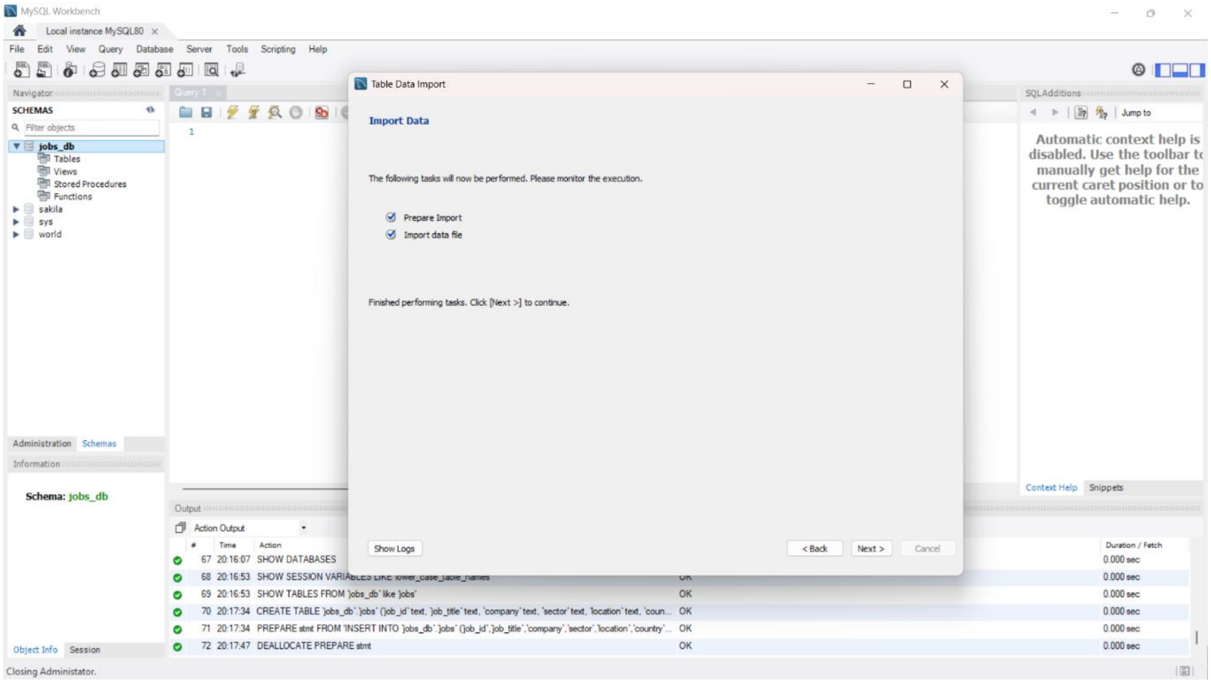
○ Prepare Import
○ Import data file

Click [Next >] to execute.

```
63 20:13:14 drop tables jobs_backup                                                0.047 sec
64 20:13:36 create table jobs            Error Code: 4028. A table must have at least one visible column.   0.016 sec
65 20:13:45 create table jobs            Error Code: 4028. A table must have at least one visible column.   0.000 sec
66 20:16:07 SHOW SESSION VARIABLES LIKE 'lower_case_table_names'    OK          0.000 sec
67 20:16:07 SHOW DATABASES                                          OK          0.000 sec
68 20:16:53 SHOW SESSION VARIABLES LIKE 'lower_case_table_names'    OK          0.000 sec
```

MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
jobs_db
Tables
Views
Stored Procedures
Functions
sakila
sys
world
Administration Schemas
Information
Schema: jobs_db
Object Info Session
Closing Administrator.
Query 1
Table Data Import
Show Logs
< Back
Next >
Cancel
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Output
Action Output
# Time Action
Duration / Fetch

8) Click Next as the file import is done.



Import Data

The following tasks will now be performed. Please monitor the execution.

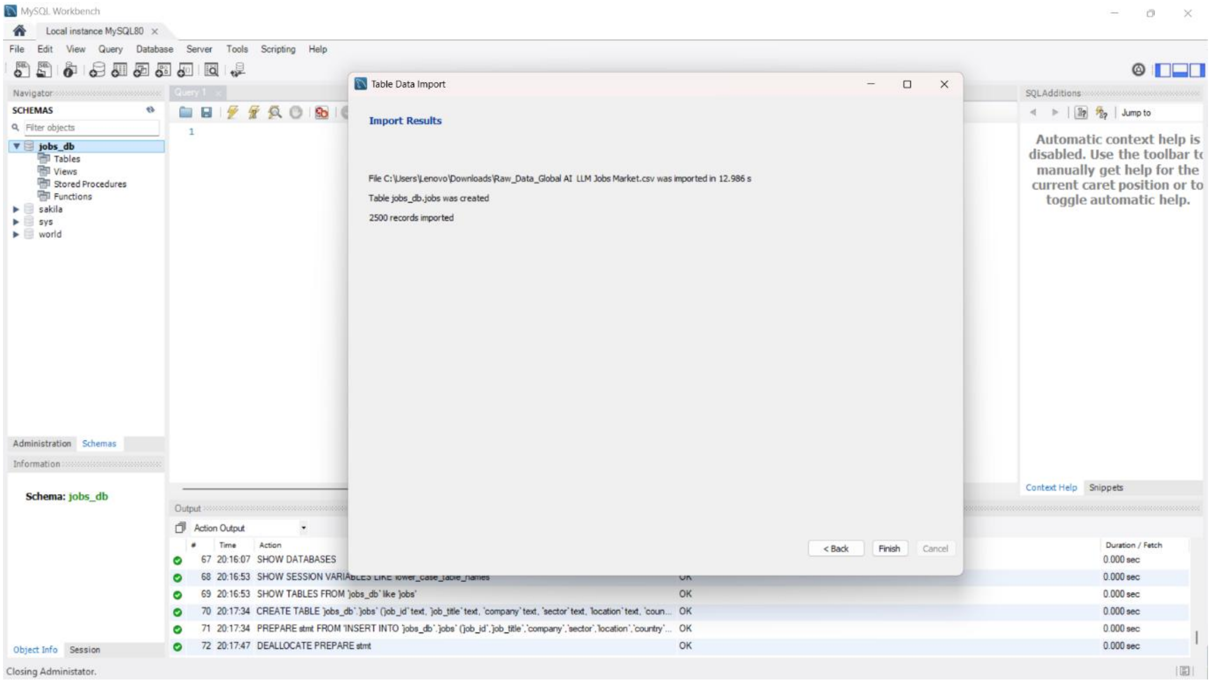
☑ Prepare Import
☑ Import data file

Finished performing tasks. Click [Next >] to continue.

```
67 20:16:07 SHOW DATABASES                                                                      0.000 sec
68 20:16:53 SHOW SESSION VARIABLES LIKE 'lower_case_table_names'                          OK    0.000 sec
69 20:16:53 SHOW TABLES FROM `jobs_db` like 'jobs'                                        OK    0.000 sec
70 20:17:34 CREATE TABLE `jobs_db`.`jobs` (`job_id` text, `job_title` text, `company` text, `sector` text, `location` text, `coun...  OK    0.000 sec
71 20:17:34 PREPARE stmt FROM 'INSERT INTO `jobs_db`.`jobs` (`job_id`,`job_title`,`company`,`sector`,`location`,`country`...  OK    0.000 sec
72 20:17:47 DEALLOCATE PREPARE stmt                                                        OK    0.000 sec
```

MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
jobs_db
Tables
Views
Stored Procedures
Functions
sakila
sys
world
Administration Schemas
Information
Schema: jobs_db
Object Info Session
Closing Administrator.
Query 1
Table Data Import
Show Logs
< Back
Next >
Cancel
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Output
Action Output
# Time Action
Duration / Fetch

9) Click finish as it shows that a table with name ‘jobs’ is created and 2500 records were imported.



MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
jobs_db
Tables
Views
Stored Procedures
Functions
sakila
sys
world
Administration Schemas
Information
Schema: jobs_db
Object Info Session
Table Data Import
< Back
Finish
Cancel
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Closing Administator.

Import Results

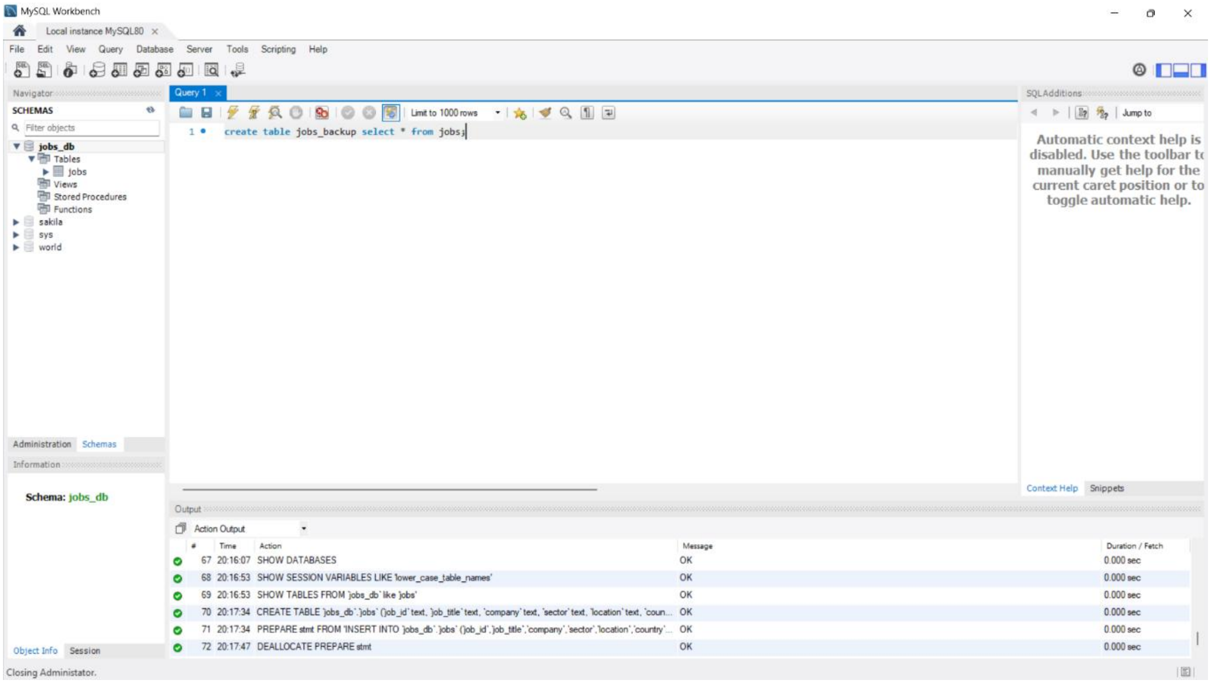
File C:\Users\Lenovo\Downloads\Raw_Data_Global AI LLM Jobs Market.csv was imported in 12.986 s

Table jobs_db.jobs was created

2500 records imported

```
Output
Action Output
  #  Time      Action                                                                                                Message  Duration / Fetch
 67  20:16:07  SHOW DATABASES                                                                                        OK       0.000 sec
 68  20:16:53  SHOW SESSION VARIABLES LIKE 'lower_case_table_names'                                                  OK       0.000 sec
 69  20:16:53  SHOW TABLES FROM `jobs_db` like 'jobs'                                                                OK       0.000 sec
 70  20:17:34  CREATE TABLE `jobs_db`.`jobs` (`job_id` text, `job_title` text, `company` text, `sector` text, `location` text, `coun...  OK  0.000 sec
 71  20:17:34  PREPARE stmt FROM 'INSERT INTO `jobs_db`.`jobs` (`job_id`,`job_title`,`company`,`sector`,`location`,`country`...  OK  0.000 sec
 72  20:17:47  DEALLOCATE PREPARE stmt                                                                               OK       0.000 sec
```

10) Create a backup of the table for a safety.



MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
jobs_db
Tables
jobs
Views
Stored Procedures
Functions
sakila
sys
world
Administration Schemas
Information
Schema: jobs_db
Object Info Session
Query 1
Limit to 1000 rows
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Closing Administator.

```sql
create table jobs_backup select * from jobs;
```

```
Output
Action Output
  #  Time      Action                                                                                                Message  Duration / Fetch
 67  20:16:07  SHOW DATABASES                                                                                        OK       0.000 sec
 68  20:16:53  SHOW SESSION VARIABLES LIKE 'lower_case_table_names'                                                  OK       0.000 sec
 69  20:16:53  SHOW TABLES FROM `jobs_db` like 'jobs'                                                                OK       0.000 sec
 70  20:17:34  CREATE TABLE `jobs_db`.`jobs` (`job_id` text, `job_title` text, `company` text, `sector` text, `location` text, `coun...  OK  0.000 sec
 71  20:17:34  PREPARE stmt FROM 'INSERT INTO `jobs_db`.`jobs` (`job_id`,`job_title`,`company`,`sector`,`location`,`country`...  OK  0.000 sec
 72  20:17:47  DEALLOCATE PREPARE stmt                                                                               OK       0.000 sec
```

11) To find missing value in MySQL

```
select
sum(job_id is null or job_id="") as missing_job_id,
sum(job_title is null or job_title="") as missing_job_tittle,
sum(company is null or company="") as missing_company,
sum(sector is null or sector="") as missing_sector,
sum(location is null or location="") as missing_location,
sum(country is null or country="") as missing_country,
sum(remote_type is null or remote_type="") as missing_remote_type,
sum(experience_level is null or experience_level ="") as missing_experience_level,
sum(salary_usd_annual is null or salary_usd_annual ="") as missing_salary_usd_annual,
sum(required_skills is null or required_skills="") as missing_required_skills,
sum(primary_llm_mentioned is null or primary_llm_mentioned = "") as
missing_primary_llm_mentioned,
sum(education_required is null or education_required="") as missing_education_required,
sum(applies_count is null or applies_count="") as missing_applies_count,
sum(visa_sponsorship is null or visa_sponsorship="") as missing_visa_sponsorship,
sum(years_experience_min is null or years_experience_min="") as missing_years_experience_min,
sum(company_size is null or company_size="") as missing_company_size,
sum(is_contract is null or is_contract="") as missing_is_contract,sum(has_equity is null or
has_equity="") as missing_has_equity
from jobs;
```

12) For Duplicate

```
select
job_id, job_title,company,sector, location, country, remote_type, experience_level,
salary_usd_annual, required_skills, primary_llm_mentioned, education_required,
count(*) as duplicate_count
from jobs
group by job_id, job_title, company, sector, location, country, remote_type, experience_level,
salary_usd_annual, required_skills,primary_llm_mentioned, education_required
having count(*) > 1;
```

13) 'TX','IL','NY','CA','WA' change to United States and ‘UK’ to United Kingdom

```
SET SQL_SAFE_UPDATES = 0;
update jobs
set country = 'United States'
where country in ('TX','IL','NY','CA','WA');
SET SQL_SAFE_UPDATES = 1;
SET SQL_SAFE_UPDATES = 0;
update jobs
set country = 'United Kingdom'
where country = 'UK';
SET SQL_SAFE_UPDATES = 1;
```

Check all distinct type

```
select distinct country
from jobs
order by country;
```

14) Tech to Technology

```
SET SQL_SAFE_UPDATES = 0;
update jobs
set sector = ‘Technology’
where country =’Tech’;
SET SQL_SAFE_UPDATES = 1;
```

15) Str to Date

```
SET SQL_SAFE_UPDATES = 0;
update jobs
set date_posted= str_to_date(date_posted,'%Y-%m-%d');
SET SQL_SAFE_UPDATES = 1;
alter table jobs
modify column date_posted date;
```

16) Modifying salary from int to decimal

```
alter table jobs
modify column salary_usd_annual decimal(12,2);
```

17) Splitting the experience level

```
alter table jobs
add column experience_years varchar(10);
```

```
SET SQL_SAFE_UPDATES = 0;
update jobs
set experience_years=case
when experience_level like 'Entry%' then '0-2'
when experience_level like 'Mid%' then '2-5'
when experience_level like 'Senior%' then '5-8'
when experience_level like 'Staff/Principal%' then '8+'
else null
end;
SET SQL_SAFE_UPDATES = 1;
```

```
SET SQL_SAFE_UPDATES = 0;
update jobs
set experience_level=case
when experience_level like 'Entry%' then 'Entry'
when experience_level like 'Mid%' then 'Mid'
when experience_level like 'Senior%' then 'Senior'
when experience_level like 'Staff/Principal%' then 'Staff/Principal'
else null
end;
SET SQL_SAFE_UPDATES = 1;
```

18) Fixing the company size

```
SET SQL_SAFE_UPDATES = 0;
update jobs
set company_size='1-50'
where company_size='Jan-50';
SET SQL_SAFE_UPDATES = 1;
```

19) None llm fixed & changing data type

```
SET SQL_SAFE_UPDATES = 0;
update jobs
set primary_llm_mentioned='No_LLM_Required'
where company_size='None';
SET SQL_SAFE_UPDATES = 1;

alter table jobs
add column primary_llm_clean varchar(25);
```

20) Dropping field (job_description_length not required)

```
alter table jobs
drop column job_description_length;
```

21) To change data type

```
alter table jobs
modify job_id varchar(15),
modify job_title varchar(50),
modify company varchar(25),
modify sector varchar(15),
modify country varchar(50),
modify remote_type varchar(15),
modify experience_level varchar(15),
modify salary_usd_annual decimal(12,2),
modify required_skills text,
modify primary_llm_mentioned varchar(25),
modify education_required varchar(25),
modify date_posted date,
modify applies_count int,
modify visa_sponsorship varchar(10),
modify years_experience_min int,
modify company_size varchar(30),
modify is_contract varchar(15),
modify has_equity varchar(5),
modify experience_years varchar(5);
```

22) Add Primary key

```
alter table jobs
add primary key(job_id);
```